

50Mhz-260Mhz 1.2V 0.13um Delay Lock Loop (DLL)

Description

G2S6004 is a general purpose clock generation DLL. The input frequency range can be from 50Mhz to 260Mhz. The total adjustment steps are 8Bits. The DLL can generate precise delays that can be programmed from 0 to 360 degrees of the reference period. This DLL can deliver optimal jitter performance over a wide frequency range. Bias block and loop filter block are all integrated to provide a compact solution. This DLL has less than 20ps RMS jitter with a compact area of 0.12 mm². The design is in TSMC 0.13um 1.2V/3.3V process.

Features

- Uses single 1.2V power supply
- Fully integrated DLL including loop filter
- Input frequency 50Mhz to 260Mhz
- Delay range 0-360 degrees of input reference
- Number of control steps 8Bits
- Reset pulse width 2 input clocks period
- Optimized PLL loop bandwidth for best jitter performance
- RMS jitter less than 20ps
- Supply voltage 1.2v
- Low Power operation of only 12mA (14.4mW)
- Power down mode with reset control
- Compact silicon area of only 0.12mm²
- TSMC 0.13um process

Applications

General purpose clock generation
 Clock synthesizer
 De-skew clock buffer

